

## 3.F Limits and Ext

It often happens that one has a CW complex  $X$  expressed as a union of an increasing sequence of subcomplexes  $X_0 \subset X_1 \subset X_2 \subset \cdots$ . For example,  $X_i$  could be the  $i$ -skeleton of  $X$ , or the  $X_i$ 's could be finite complexes whose union is  $X$ . In situations of this sort, Proposition 3.33 says that  $H_n(X; G)$  is the direct limit  $\varinjlim H_n(X_i; G)$ . Our goal in this section is to show this holds more generally for any homology theory, and to derive the corresponding formula for cohomology theories, which is a bit more complicated even for ordinary cohomology with  $\mathbb{Z}$  coefficients. For ordinary homology and cohomology the results apply somewhat more generally than just to CW complexes, since if a space  $X$  is the union of an increasing sequence of subspaces  $X_i$  with the property that each compact set in  $X$  is contained in some  $X_i$ , then the singular complex of  $X$  is the union of the singular complexes of the  $X_i$ 's, and so this gives a reduction to the CW case.

Passing to limits can often result in nonfinitely generated homology and cohomology groups. At the end of this section we describe some of the rather subtle behavior of Ext for nonfinitely generated groups.

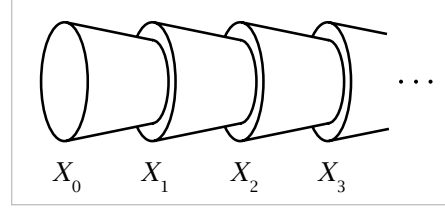
### Direct and Inverse Limits

As a special case of the general definition in §3.3, the direct limit  $\varinjlim G_i$  of a sequence of homomorphisms of abelian groups  $G_1 \xrightarrow{\alpha_1} G_2 \xrightarrow{\alpha_2} G_3 \rightarrow \cdots$  is defined to be the quotient of the direct sum  $\bigoplus_i G_i$  by the subgroup consisting of elements of the form  $(g_1, g_2 - \alpha_1(g_1), g_3 - \alpha_2(g_2), \dots)$ . It is easy to see from this definition that every element of  $\varinjlim G_i$  is represented by an element  $g_i \in G_i$  for some  $i$ , and two such representatives  $g_i \in G_i$  and  $g_j \in G_j$  define the same element of  $\varinjlim G_i$  iff they have the same image in some  $G_k$  under the appropriate composition of  $\alpha_\ell$ 's. If all the  $\alpha_i$ 's are injective and are viewed as inclusions of subgroups,  $\varinjlim G_i$  is just  $\bigcup_i G_i$ .

**Example 3F.1.** For a prime  $p$ , consider the sequence  $\mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \cdots$  with all maps multiplication by  $p$ . Then  $\varinjlim G_i$  can be identified with the subgroup  $\mathbb{Z}[1/p]$  of  $\mathbb{Q}$  consisting of rational numbers with denominator a power of  $p$ . More generally, we can realize any subgroup of  $\mathbb{Q}$  as the direct limit of a sequence  $\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \cdots$  with an appropriate choice of maps. For example, if the  $n^{\text{th}}$  map is multiplication by  $n$ , then the direct limit is  $\mathbb{Q}$  itself.

**Example 3F.2.** The sequence of injections  $\mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_{p^2} \xrightarrow{p} \mathbb{Z}_{p^3} \rightarrow \cdots$ , with  $p$  prime, has direct limit a group we denote  $\mathbb{Z}_{p^\infty}$ . This is isomorphic to  $\mathbb{Z}[1/p]/\mathbb{Z}$ , the subgroup of  $\mathbb{Q}/\mathbb{Z}$  represented by fractions with denominator a power of  $p$ . In fact  $\mathbb{Q}/\mathbb{Z}$  is isomorphic to the direct sum of the subgroups  $\mathbb{Z}[1/p]/\mathbb{Z} \approx \mathbb{Z}_{p^\infty}$  for all primes  $p$ . It is not hard to determine all the subgroups of  $\mathbb{Q}/\mathbb{Z}$  and see that each one can be realized as a direct limit of finite cyclic groups with injective maps between them. Conversely, every such direct limit is isomorphic to a subgroup of  $\mathbb{Q}/\mathbb{Z}$ .

We can realize these algebraic examples topologically by the following construction. The **mapping telescope** of a sequence of maps  $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \rightarrow \dots$  is the union of the mapping cylinders  $M_{f_i}$  with the copies of  $X_i$  in  $M_{f_i}$  and  $M_{f_{i-1}}$  identified for all  $i$ . Thus the mapping telescope is the quotient space of the disjoint union  $\coprod_i (X_i \times [i, i+1])$  in which each point  $(x_i, i+1) \in X_i \times [i, i+1]$  is identified with  $(f_i(x_i), i+1) \in X_{i+1} \times [i+1, i+2]$ . In the mapping telescope  $T$ , let  $T_i$  be the union of the first  $i$  mapping cylinders. This deformation retracts onto  $X_i$  by deformation retracting each mapping cylinder onto its right end in turn. If the maps  $f_i$  are cellular, each mapping cylinder is a CW complex and the telescope  $T$  is the increasing union of the subcomplexes  $T_i \simeq X_i$ . Then Proposition 3.33, or Theorem 3F.8 below, implies that  $H_n(T; G) \approx \varinjlim H_n(X_i; G)$ .



**Example 3F.3.** Suppose each  $f_i$  is a map  $S^n \rightarrow S^n$  of degree  $p$  for a fixed prime  $p$ . Then  $H_n(T)$  is the direct limit of the sequence  $\mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \dots$  considered in Example 3F.1 above, and  $\tilde{H}_k(T) = 0$  for  $k \neq n$ , so  $T$  is a Moore space  $M(\mathbb{Z}[1/p], n)$ .

**Example 3F.4.** In the preceding example, if we attach a cell  $e^{n+1}$  to the first  $S^n$  in  $T$  via the identity map of  $S^n$ , we obtain a space  $X$  which is a Moore space  $M(\mathbb{Z}_{p^\infty}, n)$  since  $X$  is the union of its subspaces  $X_i = T_i \cup e^{n+1}$ , which are  $M(\mathbb{Z}_{p^i}, n)$ 's, and the inclusion  $X_i \subset X_{i+1}$  induces the inclusion  $\mathbb{Z}_{p^i} \subset \mathbb{Z}_{p^{i+1}}$  on  $H_n$ .

Generalizing these two examples, we can obtain Moore spaces  $M(G, n)$  for arbitrary subgroups  $G$  of  $\mathbb{Q}$  or  $\mathbb{Q}/\mathbb{Z}$  by choosing maps  $f_i: S^n \rightarrow S^n$  of suitable degrees.

The behavior of cohomology groups is more complicated. If  $X$  is the increasing union of subcomplexes  $X_i$ , then the cohomology groups  $H^n(X_i; G)$ , for fixed  $n$  and  $G$ , form a sequence of homomorphisms

$$\dots \longrightarrow G_2 \xrightarrow{\alpha_2} G_1 \xrightarrow{\alpha_1} G_0$$

Given such a sequence of group homomorphisms, the **inverse limit**  $\varprojlim G_i$  is defined to be the subgroup of  $\prod_i G_i$  consisting of sequences  $(g_i)$  with  $\alpha_i(g_i) = g_{i-1}$  for all  $i$ . There is a natural map  $\lambda: H^n(X; G) \rightarrow \varprojlim H^n(X_i; G)$  sending an element of  $H^n(X; G)$  to its sequence of images in  $H^n(X_i; G)$  under the maps  $H^n(X; G) \rightarrow H^n(X_i; G)$  induced by inclusion. One might hope that  $\lambda$  is an isomorphism, but this is not true in general, as we shall see. However, for some choices of  $G$  it is:

**Proposition 3F.5.** *If the CW complex  $X$  is the union of an increasing sequence of subcomplexes  $X_i$  and if  $G$  is one of the fields  $\mathbb{Q}$  or  $\mathbb{Z}_p$ , then  $\lambda: H^n(X; G) \rightarrow \varprojlim H^n(X_i; G)$  is an isomorphism for all  $n$ .*

**Proof:** First we have an easy algebraic fact: Given a sequence of homomorphisms of abelian groups  $G_1 \xrightarrow{\alpha_1} G_2 \xrightarrow{\alpha_2} G_3 \rightarrow \dots$ , then  $\text{Hom}(\varprojlim G_i, G) = \varprojlim \text{Hom}(G_i, G)$

for any  $G$ . Namely, it follows from the definition of  $\varinjlim G_i$  that a homomorphism  $\varphi: \varinjlim G_i \rightarrow G$  is the same thing as a sequence of homomorphisms  $\varphi_i: G_i \rightarrow G$  with  $\varphi_i = \varphi_{i+1} \alpha_i$  for all  $i$ . Such a sequence  $(\varphi_i)$  is exactly an element of  $\varinjlim \text{Hom}(G_i, G)$ .

Now if  $G$  is a field  $\mathbb{Q}$  or  $\mathbb{Z}_p$  we have

$$\begin{aligned} H^n(X; G) &= \text{Hom}(H_n(X; G), G) \\ &= \text{Hom}(\varinjlim H_n(X_i; G), G) \\ &= \varinjlim \text{Hom}(H_n(X_i; G), G) \\ &= \varinjlim H^n(X_i; G) \end{aligned} \quad \square$$

Let us analyze what happens for cohomology with an arbitrary coefficient group, or more generally for any cohomology theory. Given a sequence of homomorphisms of abelian groups

$$\cdots \longrightarrow G_2 \xrightarrow{\alpha_2} G_1 \xrightarrow{\alpha_1} G_0$$

define a map  $\delta: \prod_i G_i \rightarrow \prod_i G_i$  by  $\delta(\cdots, g_i, \cdots) = (\cdots, g_i - \alpha_{i+1}(g_{i+1}), \cdots)$ , so that  $\varinjlim G_i$  is the kernel of  $\delta$ . Denoting the cokernel of  $\delta$  by  $\varinjlim^1 G_i$ , we have then an exact sequence

$$0 \longrightarrow \varinjlim G_i \longrightarrow \prod_i G_i \xrightarrow{\delta} \prod_i G_i \longrightarrow \varinjlim^1 G_i \longrightarrow 0$$

This may be compared with the corresponding situation for the direct limit of a sequence  $G_1 \xrightarrow{\alpha_1} G_2 \xrightarrow{\alpha_2} G_3 \longrightarrow \cdots$ . In this case one has a short exact sequence

$$0 \longrightarrow \bigoplus_i G_i \xrightarrow{\delta} \bigoplus_i G_i \longrightarrow \varinjlim G_i \longrightarrow 0$$

where  $\delta(\cdots, g_i, \cdots) = (\cdots, g_i - \alpha_{i-1}(g_{i-1}), \cdots)$ , so  $\delta$  is injective and there is no term  $\varinjlim^1 G_i$  analogous to  $\varinjlim^1 G_i$ .

Here are a few simple observations about  $\varinjlim$  and  $\varinjlim^1$ :

- If all the  $\alpha_i$ 's are isomorphisms then  $\varinjlim G_i \approx G_0$  and  $\varinjlim^1 G_i = 0$ . In fact,  $\varinjlim^1 G_i = 0$  if each  $\alpha_i$  is surjective, for to realize a given element  $(h_i) \in \prod_i G_i$  as  $\delta(g_i)$  we can take  $g_0 = 0$  and then solve  $\alpha_1(g_1) = -h_0$ ,  $\alpha_2(g_2) = g_1 - h_1$ ,  $\cdots$ .
- If all the  $\alpha_i$ 's are zero then  $\varinjlim G_i = \varinjlim^1 G_i = 0$ .
- Deleting a finite number of terms from the end of the sequence  $\cdots \rightarrow G_1 \rightarrow G_0$  does not affect  $\varinjlim G_i$  or  $\varinjlim^1 G_i$ . More generally,  $\varinjlim G_i$  and  $\varinjlim^1 G_i$  are unchanged if we replace the sequence  $\cdots \rightarrow G_1 \rightarrow G_0$  by a subsequence, with the appropriate compositions of  $\alpha_j$ 's as the maps.

**Example 3F.6.** Consider the sequence of natural surjections  $\cdots \rightarrow \mathbb{Z}_{p^3} \rightarrow \mathbb{Z}_{p^2} \rightarrow \mathbb{Z}_p$  with  $p$  a prime. The inverse limit of this sequence is a famous object in number theory, called the  **$p$ -adic integers**. Our notation for it will be  $\hat{\mathbb{Z}}_p$ . It is actually a commutative ring, not just a group, since the projections  $\mathbb{Z}_{p^{i+1}} \rightarrow \mathbb{Z}_{p^i}$  are ring homomorphisms, but we will be interested only in the additive group structure. Elements of  $\hat{\mathbb{Z}}_p$  are infinite sequences  $(\cdots, a_2, a_1)$  with  $a_i \in \mathbb{Z}_{p^i}$  such that  $a_i$  is the mod  $p^i$  reduction of  $a_{i+1}$ .

For each choice of  $a_i$  there are exactly  $p$  choices for  $a_{i+1}$ , so  $\hat{\mathbb{Z}}_p$  is uncountable. There is a natural inclusion  $\mathbb{Z} \subset \hat{\mathbb{Z}}_p$  as the constant sequences  $a_i = n \in \mathbb{Z}$ . It is easy to see that  $\hat{\mathbb{Z}}_p$  is torsionfree by checking that it has no elements of prime order.

There is another way of looking at  $\hat{\mathbb{Z}}_p$ . An element of  $\hat{\mathbb{Z}}_p$  has a unique representation as a sequence  $(\dots, a_2, a_1)$  of integers  $a_i$  with  $0 \leq a_i < p^i$  for each  $i$ . We can write each  $a_i$  uniquely in the form  $b_{i-1}p^{i-1} + \dots + b_1p + b_0$  with  $0 \leq b_j < p$ . The fact that  $a_{i+1}$  reduces mod  $p^i$  to  $a_i$  means that the numbers  $b_j$  depend only on the element  $(\dots, a_2, a_1) \in \hat{\mathbb{Z}}_p$ , so we can view the elements of  $\hat{\mathbb{Z}}_p$  as the 'base  $p$  infinite numbers'  $\dots b_1b_0$  with  $0 \leq b_i < p$  for all  $i$ , with the familiar rule for addition in base  $p$  notation. The finite expressions  $b_n \dots b_1b_0$  represent the nonnegative integers, but negative integers have infinite expansions. For example,  $-1$  has  $b_i = p - 1$  for all  $i$ , as one can see by adding 1 to this number.

Since the maps  $\mathbb{Z}_{p^{i+1}} \rightarrow \mathbb{Z}_{p^i}$  are surjective,  $\varprojlim^1 \mathbb{Z}_{p^i} = 0$ . The next example shows how  $p$ -adic integers can also give rise to a nonvanishing  $\varprojlim^1$  term.

**Example 3F.7.** Consider the sequence  $\dots \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \mathbb{Z}$  for  $p$  prime. In this case the inverse limit is zero since a nonzero integer can only be divided by  $p$  finitely often. The  $\varprojlim^1$  term is the cokernel of the map  $\delta: \prod_{\infty} \mathbb{Z} \rightarrow \prod_{\infty} \mathbb{Z}$  given by  $\delta(\gamma_1, \gamma_2, \dots) = (\gamma_1 - p\gamma_2, \gamma_2 - p\gamma_3, \dots)$ . We claim that the map  $\hat{\mathbb{Z}}_p/\mathbb{Z} \rightarrow \text{Coker } \delta$  sending a  $p$ -adic number  $\dots b_1b_0$  as in the preceding example to  $(b_0, b_1, \dots)$  is an isomorphism. To see this, note that the image of  $\delta$  consists of the sums  $\gamma_1(1, 0, \dots) + \gamma_2(-p, 1, 0, \dots) + \gamma_3(0, -p, 1, 0, \dots) + \dots$ . The terms after  $\gamma_1(1, 0, \dots)$  give exactly the relations that hold among the  $p$ -adic numbers  $\dots b_1b_0$ , and in particular allow one to reduce an arbitrary sequence  $(b_0, b_1, \dots)$  to a unique sequence with  $0 \leq b_i < p$  for all  $i$ . The term  $\gamma_1(1, 0, \dots)$  corresponds to the subgroup  $\mathbb{Z} \subset \hat{\mathbb{Z}}_p$ .

We come now to the main result of this section:

**Theorem 3F.8.** *For a CW complex  $X$  which is the union of an increasing sequence of subcomplexes  $X_0 \subset X_1 \subset \dots$  there is an exact sequence*

$$0 \rightarrow \varprojlim^1 h^{n-1}(X_i) \rightarrow h^n(X) \xrightarrow{\lambda} \varprojlim h^n(X_i) \rightarrow 0$$

*where  $h^*$  is any reduced or unreduced cohomology theory. For any homology theory  $h_*$ , reduced or unreduced, the natural maps  $\varprojlim h_n(X_i) \rightarrow h_n(X)$  are isomorphisms.*

**Proof:** Let  $T$  be the mapping telescope of the inclusion sequence  $X_0 \hookrightarrow X_1 \hookrightarrow \dots$ . This is a subcomplex of  $X \times [0, \infty)$  when  $[0, \infty)$  is given the CW structure with the integer points as 0-cells. We have  $T \simeq X$  since  $T$  is a deformation retract of  $X \times [0, \infty)$ , as we showed in the proof of Lemma 2.34 in the special case that  $X_i$  is the  $i$ -skeleton of  $X$ , but the argument works just as well for arbitrary subcomplexes  $X_i$ .

Let  $T_1 \subset T$  be the union of the products  $X_i \times [i, i+1]$  for  $i$  odd, and let  $T_2$  be the corresponding union for  $i$  even. Thus  $T_1 \cap T_2 = \coprod_i X_i$  and  $T_1 \cup T_2 = T$ . For an unreduced cohomology theory  $h^*$  we have then a Mayer-Vietoris sequence

$$\begin{array}{ccccccc}
h^{n-1}(T_1) \oplus h^{n-1}(T_2) & \longrightarrow & h^{n-1}(T_1 \cap T_2) & \longrightarrow & h^n(T) & \longrightarrow & h^n(T_1) \oplus h^n(T_2) \longrightarrow h^n(T_1 \cap T_2) \\
\wr & & \wr & & \wr & & \wr \\
\prod_i h^{n-1}(X_i) & \xrightarrow{\varphi} & \prod_i h^{n-1}(X_i) & \longrightarrow & h^n(X) & \longrightarrow & \prod_i h^n(X_i) \xrightarrow{\varphi} \prod_i h^n(X_i)
\end{array}$$

The maps  $\varphi$  making the diagram commute are given by the formula  $\varphi(\cdots, g_i, \cdots) = (\cdots, (-1)^{i-1}(g_i - \rho(g_{i+1})), \cdots)$ , the  $\rho$ 's being the appropriate restriction maps. This differs from  $\delta$  only in the sign of its even coordinates, so if we change the isomorphism  $h^k(T_1 \cap T_2) \approx \prod_i h^k(X_i)$  by inserting a minus sign in the even coordinates, we can replace  $\varphi$  by  $\delta$  in the second row of the diagram. This row then yields a short exact sequence  $0 \rightarrow \text{Coker } \delta \rightarrow h^n(X; G) \rightarrow \text{Ker } \delta \rightarrow 0$ , finishing the proof for unreduced cohomology.

The same argument works for reduced cohomology if we use the reduced telescope obtained from  $T$  by collapsing  $\{x_0\} \times [0, \infty)$  to a point, for  $x_0$  a basepoint 0-cell of  $X_0$ . Then  $T_1 \cap T_2 = \bigvee_i X_i$  rather than  $\coprod_i X_i$ , and the rest of the argument goes through unchanged. The proof also applies for homology theories, with direct products replaced by direct sums in the second row of the diagram. As we noted earlier,  $\text{Ker } \delta = 0$  in the direct limit case, and  $\text{Coker } \delta = \varinjlim$ .  $\square$

**Example 3F.9.** As in Example 3F.3, consider the mapping telescope  $T$  for the sequence of degree  $p$  maps  $S^n \rightarrow S^n \rightarrow \cdots$ . Letting  $T_i$  be the union of the first  $i$  mapping cylinders in the telescope, the inclusions  $T_1 \hookrightarrow T_2 \hookrightarrow \cdots$  induce on  $H^n(-; \mathbb{Z})$  the sequence  $\cdots \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z}$  in Example 3F.7. From the theorem we deduce that  $H^{n+1}(T; \mathbb{Z}) \approx \hat{\mathbb{Z}}_p / \mathbb{Z}$  and  $\tilde{H}^k(T; \mathbb{Z}) = 0$  for  $k \neq n+1$ . Thus we have the rather strange situation that the CW complex  $T$  is the union of subcomplexes  $T_i$  each having cohomology consisting only of a  $\mathbb{Z}$  in dimension  $n$ , but  $T$  itself has no cohomology in dimension  $n$  and instead has a huge uncountable group  $\hat{\mathbb{Z}}_p / \mathbb{Z}$  in dimension  $n+1$ . This contrasts sharply with what happens for homology, where the groups  $H_n(T_i) \approx \mathbb{Z}$  fit together nicely to give  $H_n(T) \approx \mathbb{Z}[1/p]$ .

**Example 3F.10.** A more reasonable behavior is exhibited if we consider the space  $X = M(\mathbb{Z}_{p^\infty}, n)$  in Example 3F.4 expressed as the union of its subspaces  $X_i$ . By the universal coefficient theorem, the reduced cohomology of  $X_i$  with  $\mathbb{Z}$  coefficients consists of a  $\mathbb{Z}_{p^i} = \text{Ext}(\mathbb{Z}_{p^i}, \mathbb{Z})$  in dimension  $n+1$ . The inclusion  $X_i \hookrightarrow X_{i+1}$  induces the inclusion  $\mathbb{Z}_{p^i} \hookrightarrow \mathbb{Z}_{p^{i+1}}$  on  $H_n$ , and on  $\text{Ext}$  this induced map is a surjection  $\mathbb{Z}_{p^{i+1}} \rightarrow \mathbb{Z}_{p^i}$  as one can see by looking at the diagram of free resolutions on the left:

$$\begin{array}{ccccccc}
0 \longrightarrow \mathbb{Z} & \xrightarrow{p^i} & \mathbb{Z} & \longrightarrow & \mathbb{Z}_{p^i} & \longrightarrow & 0 \\
& \downarrow \mathbb{1} & & \downarrow p & \downarrow p & & \\
0 \longrightarrow \mathbb{Z} & \xrightarrow{p^{i+1}} & \mathbb{Z} & \longrightarrow & \mathbb{Z}_{p^{i+1}} & \longrightarrow & 0
\end{array}
\quad
\begin{array}{ccccccc}
0 \longleftarrow \text{Ext}(\mathbb{Z}_{p^i}, \mathbb{Z}) & \longleftarrow & \text{Hom}(\mathbb{Z}, \mathbb{Z}) & \longleftarrow & \cdots \\
& \uparrow & & \uparrow \mathbb{1} & \\
0 \longleftarrow \text{Ext}(\mathbb{Z}_{p^{i+1}}, \mathbb{Z}) & \longleftarrow & \text{Hom}(\mathbb{Z}, \mathbb{Z}) & \longleftarrow & \cdots
\end{array}$$

Applying  $\text{Hom}(-, \mathbb{Z})$  to this diagram, we get the diagram on the right, with exact rows, and the left-hand vertical map is a surjection since the vertical map to the right of it is surjective. Thus the sequence  $\cdots \rightarrow H^{n+1}(X_2; \mathbb{Z}) \rightarrow H^{n+1}(X_1; \mathbb{Z})$  is the

sequence in Example 3F.6, and we deduce that  $H^{n+1}(X; \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$ , the  $p$ -adic integers, and  $\tilde{H}^k(X; \mathbb{Z}) = 0$  for  $k \neq n+1$ .

This example can be related to the preceding one. If we view  $X$  as the mapping cone of the inclusion  $S^n \hookrightarrow T$  of one end of the telescope, then the long exact sequences of homology and cohomology groups for the pair  $(T, S^n)$  reduce to the short exact sequences at the right.

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_n(S^n) & \longrightarrow & H_n(T) & \longrightarrow & H_n(X) \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ & & \mathbb{Z} & & \mathbb{Z}[1/p] & & \mathbb{Z}_{p^\infty} \\ \\ 0 & \longrightarrow & H^n(S^n) & \longrightarrow & H^{n+1}(X) & \longrightarrow & H^{n+1}(T) \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ & & \mathbb{Z} & & \hat{\mathbb{Z}}_p & & \hat{\mathbb{Z}}_p/\mathbb{Z} \end{array}$$

From these examples and the universal coefficient theorem we obtain isomorphisms  $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$  and  $\text{Ext}(\mathbb{Z}[1/p], \mathbb{Z}) \approx \hat{\mathbb{Z}}_p/\mathbb{Z}$ . These can also be derived directly from the definition of  $\text{Ext}$ . A free resolution of  $\mathbb{Z}_{p^\infty}$  is

$$0 \longrightarrow \mathbb{Z}^\infty \xrightarrow{\varphi} \mathbb{Z}^\infty \longrightarrow \mathbb{Z}_{p^\infty} \longrightarrow 0$$

where  $\mathbb{Z}^\infty$  is the direct sum of an infinite number of  $\mathbb{Z}$ 's, the sequences  $(x_1, x_2, \dots)$  of integers all but finitely many of which are zero, and  $\varphi$  sends  $(x_1, x_2, \dots)$  to  $(px_1 - x_2, px_2 - x_3, \dots)$ . We can view  $\varphi$  as the linear map corresponding to the infinite matrix with  $p$ 's on the diagonal,  $-1$ 's just above the diagonal, and  $0$ 's everywhere else. Clearly  $\text{Ker } \varphi = 0$  since integers cannot be divided by  $p$  infinitely often. The image of  $\varphi$  is generated by the vectors  $(p, 0, \dots), (-1, p, 0, \dots), (0, -1, p, 0, \dots), \dots$  so  $\text{Coker } \varphi \approx \mathbb{Z}_{p^\infty}$ . Dualizing by taking  $\text{Hom}(-, \mathbb{Z})$ , we have  $\text{Hom}(\mathbb{Z}^\infty, \mathbb{Z})$  the infinite direct product of  $\mathbb{Z}$ 's, and  $\varphi^*(y_1, y_2, \dots) = (py_1, py_2 - y_1, py_3 - y_2, \dots)$ , corresponding to the transpose of the matrix of  $\varphi$ . By definition,  $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) = \text{Coker } \varphi^*$ . The image of  $\varphi^*$  consists of the infinite sums  $y_1(p, -1, 0, \dots) + y_2(0, p, -1, 0, \dots) + \dots$ , so  $\text{Coker } \varphi^*$  can be identified with  $\hat{\mathbb{Z}}_p$  by rewriting a sequence  $(z_1, z_2, \dots)$  as the  $p$ -adic number  $\dots z_2 z_1$ .

The calculation  $\text{Ext}(\mathbb{Z}[1/p], \mathbb{Z}) \approx \hat{\mathbb{Z}}_p/\mathbb{Z}$  is quite similar. A free resolution of  $\mathbb{Z}[1/p]$  can be obtained from the free resolution of  $\mathbb{Z}_{p^\infty}$  by omitting the first column of the matrix of  $\varphi$  and, for convenience, changing sign. This gives the formula  $\varphi(x_1, x_2, \dots) = (x_1, x_2 - px_1, x_3 - px_2, \dots)$ , with the image of  $\varphi$  generated by the elements  $(1, -p, 0, \dots), (0, 1, -p, 0, \dots), \dots$ . The dual map  $\varphi^*$  is given by  $\varphi^*(y_1, y_2, \dots) = (y_1 - py_2, y_2 - py_3, \dots)$ , and this has image consisting of the sums  $y_1(1, 0, \dots) + y_2(-p, 1, 0, \dots) + y_3(0, -p, 1, 0, \dots) + \dots$ , so we get  $\text{Ext}(\mathbb{Z}[1/p], \mathbb{Z}) = \text{Coker } \varphi^* \approx \hat{\mathbb{Z}}_p/\mathbb{Z}$ . Note that  $\varphi^*$  is exactly the map  $\delta$  in Example 3F.7.

It is interesting to note also that the map  $\varphi: \mathbb{Z}^\infty \rightarrow \mathbb{Z}^\infty$  in the two cases  $\mathbb{Z}_{p^\infty}$  and  $\mathbb{Z}[1/p]$  is precisely the cellular boundary map  $H_{n+1}(X^{n+1}, X^n) \rightarrow H_n(X^n, X^{n-1})$  for the Moore space  $M(\mathbb{Z}_{p^\infty}, n)$  or  $M(\mathbb{Z}[1/p], n)$  constructed as the mapping telescope of the sequence of degree  $p$  maps  $S^n \rightarrow S^n \rightarrow \dots$ , with a cell  $e^{n+1}$  attached to the first  $S^n$  in the case of  $\mathbb{Z}_{p^\infty}$ .

### More About Ext

The functors  $\text{Hom}$  and  $\text{Ext}$  behave fairly simply for finitely generated groups, when cohomology and homology are essentially the same except for a dimension shift in the torsion. But matters are more complicated in the nonfinitely generated case. A useful tool for getting a handle on this complication is the following:

**Proposition 3F.11.** *Given an abelian group  $G$  and a short exact sequence of abelian groups  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ , there are exact sequences*

$$0 \rightarrow \text{Hom}(G, A) \rightarrow \text{Hom}(G, B) \rightarrow \text{Hom}(G, C) \rightarrow \text{Ext}(G, A) \rightarrow \text{Ext}(G, B) \rightarrow \text{Ext}(G, C) \rightarrow 0$$

$$0 \rightarrow \text{Hom}(C, G) \rightarrow \text{Hom}(B, G) \rightarrow \text{Hom}(A, G) \rightarrow \text{Ext}(C, G) \rightarrow \text{Ext}(B, G) \rightarrow \text{Ext}(A, G) \rightarrow 0$$

**Proof:** A free resolution  $0 \rightarrow F_1 \rightarrow F_0 \rightarrow G \rightarrow 0$  gives rise to a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}(F_0, A) & \longrightarrow & \text{Hom}(F_0, B) & \longrightarrow & \text{Hom}(F_0, C) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{Hom}(F_1, A) & \longrightarrow & \text{Hom}(F_1, B) & \longrightarrow & \text{Hom}(F_1, C) \longrightarrow 0 \end{array}$$

Since  $F_0$  and  $F_1$  are free, the two rows are exact, as they are simply direct products of copies of the exact sequence  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ , in view of the general fact that  $\text{Hom}(\bigoplus_i G_i, H) = \prod_i \text{Hom}(G_i, H)$ . Enlarging the diagram by zeros above and below, it becomes a short exact sequence of chain complexes, and the associated long exact sequence of homology groups is the first of the two six-term exact sequences in the proposition.

To obtain the other exact sequence we will construct the commutative diagram at the right, where the columns are free resolutions and the rows are exact. To start, let  $F_0 \rightarrow A$  and  $F_0'' \rightarrow C$  be surjections from free abelian groups onto  $A$  and  $C$ . Then let  $F_0' = F_0 \oplus F_0''$ , with the obvious

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F_1 & \longrightarrow & F_1' & \longrightarrow & F_1'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F_0 & \longrightarrow & F_0' & \longrightarrow & F_0'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

maps in the second row, inclusion and projection. The map  $F_0' \rightarrow B$  is defined on the summand  $F_0$  to make the lower left square commute, and on the summand  $F_0''$  it is defined by sending basis elements of  $F_0''$  to elements of  $B$  mapping to the images of these basis elements in  $C$ , so the lower right square also commutes. Now we have the bottom two rows of the diagram, and we can regard these two rows as a short exact sequence of two-term chain complexes. The associated long exact sequence of homology groups has six terms, the first three being the kernels of the three vertical maps to  $A$ ,  $B$ , and  $C$ , and the last three being the cokernels of these maps. Since the vertical maps to  $A$  and  $C$  are surjective, the fourth and sixth of the six homology groups vanish, hence also the fifth, which says the vertical map to  $B$  is surjective. The first three of the original six homology groups form a short exact sequence, and we let this be the top row of the diagram, formed by the kernels of the vertical maps to  $A$ ,  $B$ , and  $C$ . These kernels are subgroups of free abelian groups, hence are also free.

Thus the three columns are free resolutions. The upper two squares automatically commute, so the construction of the diagram is complete.

The first two rows of the diagram split by freeness, so applying  $\text{Hom}(-, G)$  yields a diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \text{Hom}(F_0'', G) & \longrightarrow & \text{Hom}(F_0', G) & \longrightarrow & \text{Hom}(F_0, G) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \text{Hom}(F_1'', G) & \longrightarrow & \text{Hom}(F_1', G) & \longrightarrow & \text{Hom}(F_1, G) & \longrightarrow & 0 \end{array}$$

with exact rows. Again viewing this as a short exact sequence of chain complexes, the associated long exact sequence of homology groups is the second six-term exact sequence in the statement of the proposition.  $\square$

The second sequence in the proposition says in particular that an injection  $A \rightarrow B$  induces a surjection  $\text{Ext}(B, C) \rightarrow \text{Ext}(A, C)$  for any  $C$ . For example, if  $A$  has torsion, this says  $\text{Ext}(A, \mathbb{Z})$  is nonzero since it maps onto  $\text{Ext}(\mathbb{Z}_n, \mathbb{Z}) \approx \mathbb{Z}_n$  for some  $n > 1$ . The calculation  $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$  earlier in this section shows that torsion in  $A$  does not necessarily yield torsion in  $\text{Ext}(A, \mathbb{Z})$ , however.

Also useful are the formulas

$$\text{Ext}(\bigoplus_i A_i, B) \approx \prod_i \text{Ext}(A_i, B) \qquad \text{Ext}(A, \bigoplus_i B_i) \approx \bigoplus_i \text{Ext}(A, B_i)$$

whose proofs we leave as exercises. For example, since  $\mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Z}_{p^\infty}$  we obtain  $\text{Ext}(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \approx \prod_p \hat{\mathbb{Z}}_p$  from the calculation  $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$ . Then from the exact sequence  $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$  we get  $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \approx (\prod_p \hat{\mathbb{Z}}_p)/\mathbb{Z}$  using the second exact sequence in the proposition.

In these examples the groups  $\text{Ext}(A, \mathbb{Z})$  are rather large, and the next result says this is part of a general pattern:

**Proposition 3F.12.** *If  $A$  is not finitely generated then either  $\text{Hom}(A, \mathbb{Z})$  or  $\text{Ext}(A, \mathbb{Z})$  is uncountable. Hence if  $H_n(X; \mathbb{Z})$  is not finitely generated then either  $H^n(X; \mathbb{Z})$  or  $H^{n+1}(X; \mathbb{Z})$  is uncountable.*

Both possibilities can occur, as we see from the examples  $\text{Hom}(\bigoplus_\infty \mathbb{Z}, \mathbb{Z}) \approx \prod_\infty \mathbb{Z}$  and  $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$ .

This proposition has some interesting topological consequences. First, it implies that if a space  $X$  has  $\tilde{H}^*(X; \mathbb{Z}) = 0$ , then  $\tilde{H}_*(X; \mathbb{Z}) = 0$ , since the case of finitely generated homology groups follows from our earlier results. And second, it says that one cannot always construct a space  $X$  with prescribed cohomology groups  $H^n(X; \mathbb{Z})$ , as one can for homology. For example there is no space whose only nonvanishing  $\tilde{H}^n(X; \mathbb{Z})$  is a countable nonfinitely generated group such as  $\mathbb{Q}$  or  $\mathbb{Q}/\mathbb{Z}$ . Even in the finitely generated case the dimension  $n = 1$  is somewhat special since the group  $H^1(X; \mathbb{Z}) \approx \text{Hom}(H_1(X), \mathbb{Z})$  is always torsionfree.